

⇒ Sum of odd indexed elements

$$\text{arr}() = \begin{matrix} & 0 & 1 & 2 & 3 & 4 \\ [& 2 & 4 & 3 & 1 & 5 &] \end{matrix}$$

$$\text{pfo} = [0 \quad 4 \quad 4 \quad 5 \quad 5]$$

$$\text{pfe} = [2 \quad 2 \quad 5 \quad 5 \quad 10]$$

- Given an array of size N, count the number of special index in the array.

Note: **Special Indices** are those after removing which, sum of all **EVEN** indexed elements is equal to sum of all **ODD** indexed elements.

Example

$A[] = \{ 4, 3, 2, 7, 6, -2 \}$
 Ans = 2

We can see that after removing 0th and 2nd index S_e and S_o are equal.

i	A[i]	S_e	S_o	
0	4	8	8	✓
1	3	9	8	✗
2	2	9	9	✓
3	7	4	9	✗
4	6	4	10	✗
5	-2	12	10	✗

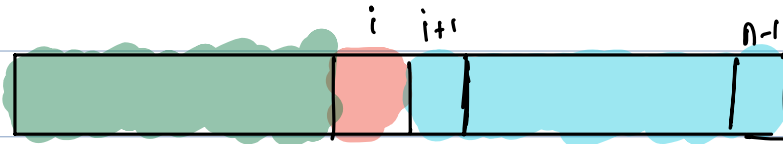
Quiz 2

$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 7 & 10 \end{bmatrix}$

$\begin{matrix} 0 & 1 & 2 & 3 \\ 4 & 1 & 7 & 10 \end{matrix}$

= 11

P_o & $P_e \Rightarrow$ you have computed.



Sum of even indexed elements

$[S_e]$

$$\Rightarrow P_e[i-1]$$

$$+ P_o[n-1] - P_o[i]$$

Sum of odd indexed elements

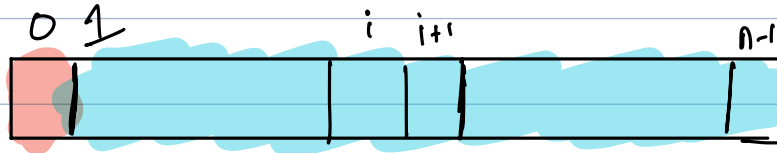
$[S_o]$

$$\Rightarrow P_o[i-1]$$

$$+ P_e[n-1] - P_e[i]$$

Edge case $i=0$

$$P_o[n-1] - P_o[0]$$



$$[S_e] = P_o[n-1] - P_o[i]$$

$$[S_o] = P_e[n-1] - P_e[i]$$

Pseudo Code

- 1) Compute Pfe .
- 2) Compute Ppo .

```
int ans = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
    int Se, So;
```

```
    if (i == 0) {
```

```
        Se =  $Ppo[n-1] - Ppo[i]$ ;
```

```
        So =  $Pfe[n-1] - Pfe[i]$ ;
```

```
    } else {
```

```
        Se =  $Pfe[i-1] + Ppo[n-1] - Ppo[i]$ ;
```

```
        So =  $Ppo[i-1] + Pfe[n-1] - Pfe[i]$ ;
```

```
    }
```

```
    if (So == Se)
```

```
        ans++;
```

```
}
```

```
return ans;
```

Q3 Given n elements in an array.
Also given q queries $[l, r]$.
Find the count of even numbers
from $[l, r]$.

arr: $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 1 & 6 & 5 & 7 & 8 & 10 & 11 \end{matrix}$

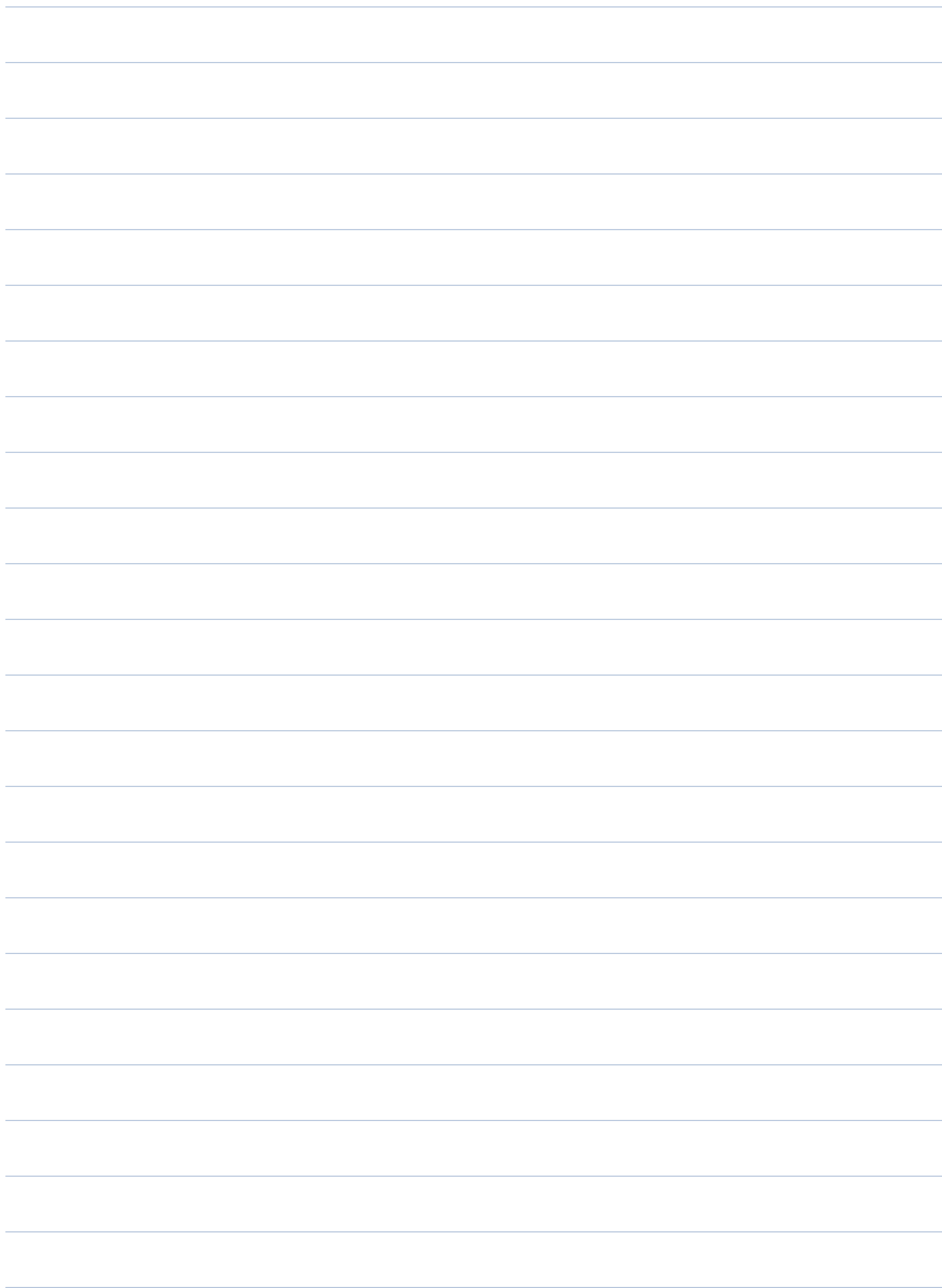
l	r	ans
1	5	2
4	7	2

arr: $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 1 & 6 & 5 & 7 & 8 & 10 & 11 \end{matrix}$



arr: $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \end{matrix}$

pf: $\begin{matrix} 1 & 1 & 2 & 2 & 2 & 3 & 4 & 4 \end{matrix}$



Q4

Problem Description

Given an integer array A containing N distinct integers, you have to find all the leaders in array A.

An element is a leader if it is strictly greater than all the elements to its right side.

NOTE:

- The rightmost element is always a leader.
- You can return the leaders in any order.

Problem Constraints

- $1 \leq N \leq 10^5$
- $1 \leq A[i] \leq 10^8$

arr =

	0	1	2	3	4	5
	16	17	4	3	5	2
	X	✓	X	X	✓	✓

⇒ [17, 5, 2]

arr =

	0	1	2	3	4	5
	16	17	4	3	5	2

max. element =

	17	17	5	5	5	2
--	----	----	---	---	---	---

leader =

	X	✓	X	X	✓	✓
--	---	---	---	---	---	---

Pseudo Code

int max_element = arr[n-1];

list<int> ans;

ans.add(arr[n-1]);

for (int i = (n-2); i ≥ 0 ; i--) {

if (arr[i] > max_element) {

ans.add(arr[i]);

max_element = arr[i];

}

return ans;

[https://notability.com/app/note/
e15ac905-dc68-45af-
b92b-4c5236278a3e](https://notability.com/app/note/e15ac905-dc68-45af-b92b-4c5236278a3e)